

Erdős Problem #625

EP-625

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

The cochromatic number of G , denoted by $\zeta(G)$, is the minimum number of colours needed to colour the vertices of G such that each colour class induces either a complete graph or empty graph. Let $\chi(G)$ denote the chromatic number.

If G is a random graph with n vertices and each edge included independently with probability $1/2$ then is it true that almost surely $\chi(G) - \zeta(G) \rightarrow \infty$ as $n \rightarrow \infty$?